When LLM Labels Fail Together

HanRui Zhang[1]
High School Senior, TOPS Program
Marc Garneau Collegiate Institute
Toronto, Canada
daniel21cn2016@gmail.com

***Abstract*—Large language models (LLMs) can label many texts in one prompt, saving repeated instructions and inference cost, but the shared response may make their errors dependent. To measure this dependence while holding visible context fixed, we use one-target controls that show the same neighboring texts in the same order. Across a batch-16 grid of four open-weight model families and four text-classification tasks, mean excess error covariance (the added tendency for errors to occur together) was 0.00961, with a one-sided randomization p-value of 0.0002. All 16 estimates were positive, and 13 exceeded the prespecified practical threshold of 0.002. Across four selected model–task combinations, mean pairwise dependence weakened after batch size 4, but total dependence raised the estimated variance from 1.08 to 2.38 times the independent-label value. A secondary analysis completed after the main experiment found that treating every label as independent made confidence intervals too narrow in three of four model–task combinations. When intervals instead accounted for which labels came from the same LLM call, they contained the target error rate about 95% of the time, as intended. Each batched annotation call should therefore be treated as a statistical cluster. Studies using batched labels should record which call and protocol produced each label and account for within-call dependence when estimating uncertainty.**

***Index Terms*—large language models, data annotation, batch prompting, dependent errors, uncertainty estimation**

I. Introduction

Large language models (LLMs) can label multiple texts in one response, reducing repeated instructions and inference cost. Prior work on batch prompting primarily measures efficiency and per-item accuracy [1], [2]. It therefore remains unclear whether a shared generation makes errors across items dependent.

Overall accuracy does not reveal whether errors are independent. When 16 texts share one prompt and response, the model can carry an interpretation or formatting decision across the call. Positively correlated errors leave a dataset with less independent information than its row count suggests. Confidence intervals that treat the labels as independent may then be too narrow even when their point estimates are unbiased. Classical methods for clustered data address this gap [3], [4].

To test whether requesting multiple labels in one call makes errors co-occur, we pair each joint prompt with an exact-context control that shows the same documents in the same order but requests only one target label (Fig. 1). The comparison therefore holds context fixed while changing whether labels share a response. The study combines a 4×4 model–task grid at batch size 16, five batch sizes from 2 to 32 in one prespecified combination per model family, and a recovered analysis comparing item-level confidence intervals with intervals based on calls.

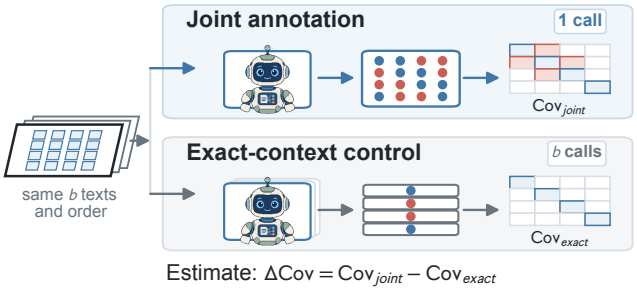


Fig. 1. Matched-context design for a batch of b texts. The joint condition returns b labels in one call, whereas the control generates each target label in one of b separate calls using the same visible context. Coral marks illustrate coupled errors and off-diagonal covariance; blue marks show other outputs and diagonal entries. Covariance matrices are schematic. The reported measure subtracts exact-context covariance from joint-call covariance. The 4×4 model–task grid and recovered coverage analysis use $b = 16$, while the batch-size study uses $b \in \{2, 4, 8, 16, 32\}$ in four prespecified model–task combinations.

The study makes two contributions. First, it measures excess error covariance across a balanced 4×4 grid and separates weaker pairwise dependence from greater total information loss as batch size grows. Second, a recovered repeated-sampling analysis measures how often 95% confidence intervals miss their target when call membership is ignored. We recommend storing a call identifier with every LLM-generated label and using it to estimate uncertainty.

II. Related Work

LLMs are increasingly evaluated as substitutes or complements for human annotators. Gilardi et al. [5] found that ChatGPT could outperform crowd workers on several text-annotation tasks, and Tan et al. [6] organize the broader literature around annotation generation, assessment, and downstream use. Larger evaluations add important qualifications. Ziems et al. [7] found substantial variation across models and computational social science tasks. Pangakis and Wolken [8] found that performance varied across 27 tasks from 11 protected datasets and argued for validation against human labels.

[1]Code: https://anonymous.4open.science/r/batched-annotation-D206/

Even minor prompt-format changes can alter model performance [9]. These studies show that LLM annotation quality depends on the task and protocol, but they mainly measure item-level accuracy, agreement, or sensitivity. Whereas Kim et al. [10] measure error correlation across distinct models, we measure dependence among items labeled within one shared generation.

Batch prompting is the closest setting to our study. Cheng et al. [1] evaluate its token, time, and accuracy tradeoffs across ten datasets. Pipal et al. [2] study annotation more directly across eight production models, four tasks, and batch sizes up to 1,000. Both focus on efficiency and per-item accuracy rather than whether labels sharing a response also share errors. Lin et al. [11] show that batch-prompt performance depends on item position and order because earlier outputs change the decoder context. Yue and Yao [12] show that a malicious instruction in one item can disrupt other outputs in the same batch. We study ordinary classification inputs and compare them with target-only calls that show the same context.

Methods for correlated data are also established outside batched annotation. Gligorić et al. [13] use LLM confidence and selective human annotation to construct confidence intervals with correct coverage. Llewellyn et al. [14] cluster semantically dependent benchmark prompts, while Zeng et al. [15] address missing responses when observations from the same user are dependent. These studies address annotation error, prompt structure, or user-level dependence. We test dependence created when multiple labels share one generation. Standard methods already handle known clusters [3], [4], [16], [17]. Our contribution is a matched estimate of call-associated error covariance and a repeated-sampling evaluation of confidence-interval coverage when call membership is discarded.

III. Experimental Design

A. Models and Tasks

The experiment crosses four similarly sized open-weight model families—Qwen3-8B, Granite-3.3-8B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 [18], [19], [20], [21]—with four tasks: toxicity classification on Civil Comments [22], natural-language inference on MultiNLI [23], sentiment classification on IMDb [24], and topic classification on AG News [25]. Each model–task combination forms a *cell*. Civil Comments scores ≥ 0.5 were labeled toxic. After fixed exclusions, we hash-ordered candidates and divided the confirmation set into 16 equal-sized text-length strata. Because dataset labels served as references, an "error" means disagreement with that label rather than an objectively wrong prediction.

Every model used the same instruction template, JSON grammar, parser, greedy decoding, and 32,768-token context limit. Before observing joint outputs, we fixed each task at 8,192 items using solo-prompt error rates and CPU-based power simulations. Neither accuracy nor parse rate was later used to remove a cell.

B. Joint and Exact-Context Conditions

Within each cell, three independent randomized assignments formed length-balanced groups of 16 items. A *joint* prompt requested all 16 labels, whereas an *exact-context* prompt showed the same items in the same order but requested only one target label. Repeating that target-only request for every item produced 16 separate calls, which kept the visible context constant while removing the shared generated response.

Let $e_i \in \{0, 1\}$ indicate disagreement with the reference label, and let $\bar{e}$ be the mean error for each model, task, condition, and assignment. After placing these fixed errors into matched 16-item groups representing either one joint call or 16 corresponding exact-context calls, we averaged $(e_i - \bar{e})(e_j - \bar{e})$ over distinct item pairs within each group and then across the three assignment contrasts. The primary statistic for each cell was

$$\Delta = \mathrm{Cov}_{\mathrm{joint}}(e_i, e_j) - \mathrm{Cov}_{\mathrm{exact}}(e_i, e_j). \tag{1}$$

A positive Δ means that errors co-occur more often when labels share a response than when they only share context. Under the *sharp null*, which holds outputs fixed and assumes that group assignment has no effect, 4,999 paired replays permuted items within assignment, length stratum, and execution block while preserving group size and repeated appearances, with both conditions using the same regrouping. From these replays, we calculated upper-tail p-values and applied Holm's adjustment across the 16 cells. The 95% randomization-based interval subtracts mean-centered replay quantiles from the estimate and applies only to the fixed outputs and replay design. Set before joint results were observed, the practical threshold $\Delta \geq 0.002$ was evaluated separately from statistical rejection. The combined estimate gives every cell equal weight and applies only to the tested models, tasks, and items.

To prevent selective parsing choices from creating the result, we retained malformed outputs under four prespecified rules: count unresolved targets as correct, count them as errors, drop only unresolved targets, or drop their entire calls.

C. Batch-Size Robustness

In one prespecified cell per model family, we repeated the joint-versus-exact comparison at batch sizes 2, 4, 8, 16, and 32. Each size used all 8,192 items, three nested randomized assignments, and the same output format. Dividing covariance by its condition-specific Bernoulli variance then produced excess within-call error correlation, which we regressed on $\log_2$ batch size for each cell before averaging the four slopes. Because items recur across assignments, the slopes are descriptive rather than independent replicates. The size-32 block-bootstrap sensitivity separately resamples assignments and breaks recurring item identity, so it does not support the main uncertainty claims. Pooled excess correlation ρ_Δ was converted into a design effect, the factor by which dependence increases variance, as $1+(b-1)\rho_\Delta$. The corresponding effective sample size, $8{,}192/[1 + (b - 1)\rho_\Delta]$, is the equivalent number of independent labels.

D. Recovered Repeated-Sampling Sensitivity

The repeated-sampling analysis tested whether call dependence changes uncertainty in the mean disagreement rate. Selected by a fixed task order and solo-prompt results before covariance was inspected, its four cells were Granite/IMDb, Qwen/Civil Comments, Llama/Civil Comments, and Mistral/MultiNLI. Each of 250 reference and 400 evaluation replicates sampled 512 item appearances and generated 32 new 16-item calls. While the reference replicates estimated the target disagreement rate, every evaluation replicate produced three nominal 95% intervals: a Wald interval treating 512 errors as independent, a t interval over 32 call means, and a whole-call bootstrap interval. Coverage is the percentage containing the target rate. A calibrated 95% method should cover it about 95% of the time. With 400 replicates, its maximum Monte Carlo standard error is 2.5 percentage points. Prespecified precision and stability checks supported assessing undercoverage for Granite, Qwen, and Mistral.

Of 83,200 calls, 1,095 (1.3%) with malformed or truncated responses were recovered. Although the reruns changed the output format and maximum length, they kept the original prompt, model, seed, numerical precision, runtime, and decoding settings, and every successful original output was retained. Because the original coverage analysis could not be completed, the recovered analysis is a secondary, post-execution sensitivity, whereas the primary covariance analysis was unchanged and includes all prespecified malformed-output rules.

E. Artifact Availability

The reviewer artifact contains analysis code, frozen protocols and prompts, dataset indices, and analysis-ready error arrays. It reproduces the reported analyses. Because it excludes dataset text, raw model responses, and cloud execution records, it cannot verify model execution or parsing. It is available at https://anonymous.4open.science/r/batched-annotation-D206/README.md.

IV. Results

A. Dependence Recurs Across Models and Tasks

The mean excess covariance across cells was 0.00961, with a 95% randomization-based interval of [0.00936, 0.00986] and one-sided replay $p = 0.0002$. All 16 cells rejected the sharp null after Holm adjustment, and 13 also met the separate 0.002 practical threshold. Those 13 cells covered all four model families and all four tasks (Fig. 2). Every mean calculated after removing one model or one task remained positive, so no single family or dataset drove the combined result.

All four prespecified rules for malformed outputs gave combined excess covariance from 0.00752 to 0.01550, always above the practical threshold. A sensitivity using 3,002 recovered calls gave 0.00790. Thirteen cells passed a combined rule requiring both a large enough effect and rejection by the replay test, although two differed from the primary set. All 48 assignment-specific contrasts were positive, and means calculated after removing one assignment ranged from 0.00935 to 0.00992. Across cells and assignments, standardized covariance averaged 0.0557 in the joint condition and 0.0014 in the exact-context condition. These values gave mean design effects of 1.835 and 1.021 and effective sample sizes of 5,022 and 8,105 per 8,192 labels, respectively.

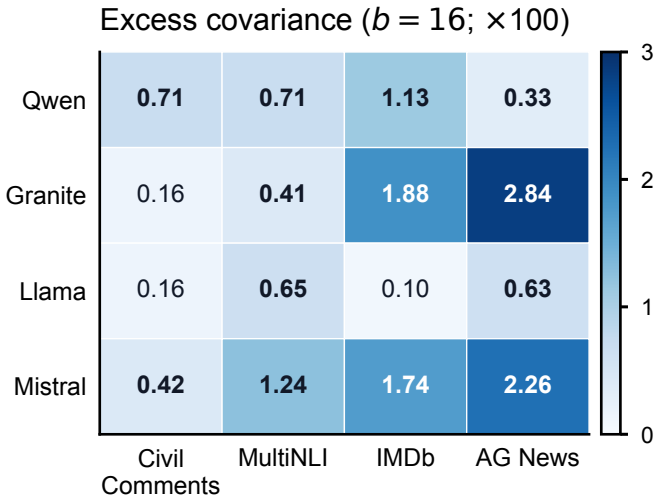


Fig. 2. Joint-minus-exact error covariance at batch size 16 (×100, 8,192 items per cell). Bold estimates meet the prespecified 0.002 practical threshold. All 16 cells reject the sharp null after Holm adjustment.

A post-hoc Civil Comments sensitivity kept four-cell covariance positive at thresholds 0.4 (0.00343), 0.5 (0.00464), and 0.6 (0.00626), and after excluding 509 ambiguous items (0.00524). Both Civil Comments batch-size slopes remained negative, and the repeated-sampling conclusions were qualitatively unchanged.

The information loss varied across settings. A descriptive calculation assuming equal correlation among all pairs within a call gave cell-level design effects from 1.14 to 2.87. For 8,192 labeled rows, these values correspond to effective sample sizes from 7,162 down to 2,850. As a negative control, fake batches assembled from separately generated solo labels had covariance between −0.00086 and 0.00082 across the 16 cells. Regrouping independent labels therefore did not reproduce the dependence observed when labels shared a response.

B. Dependence Persists but Does Not Grow Uniformly with Batch Size

Standardized excess dependence averaged across the four cells was positive at every tested size: 0.079 at batch size 2, 0.081 at size 4, and 0.045 at size 32 (Fig. 3). The mean slope was −0.00910 per doubling of batch size. It was negative in each complete assignment (−0.00960, −0.00879, and −0.00889). Dependence between pairs therefore decreased on average after size 4, although the three-assignment comparison is descriptive.

Three cells had negative slopes, whereas Granite/IMDb had a positive slope, so no single trend described every selected cell.

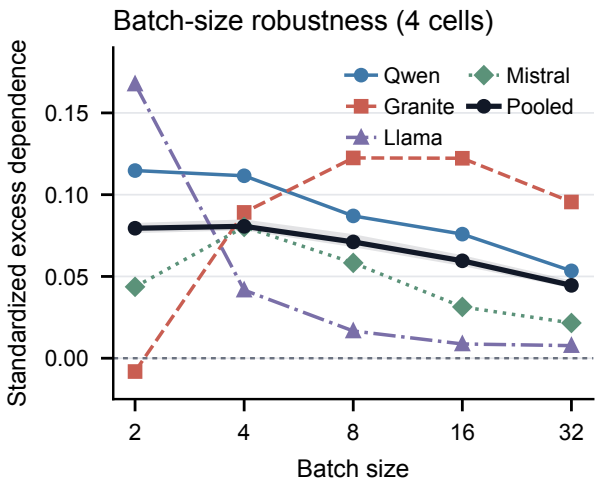


Fig. 3. Standardized joint-minus-exact dependence across batch sizes 2–32 in four prespecified cells. The black line is the equal-cell mean, and the shading spans the three complete randomized assignments.

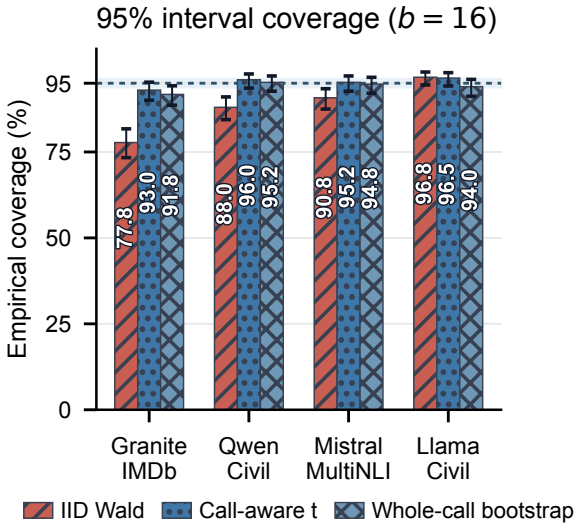


Fig. 4. Recovered post-execution analysis of coverage: the share of 400 repeated confidence intervals containing the target error rate at batch size 16. Error bars are exact 95% binomial intervals, and shading marks 94–96% coverage.

Weaker pairwise dependence did not prevent the total variance penalty from growing. The descriptive design effect increased from 1.079 at batch size 2 (assignment range 1.076–1.083) to 2.381 at size 32 (2.328–2.473). Relative to the exact-context baseline, the corresponding effective sample size fell from 7,589 to 3,440 out of 8,192 rows. Each pair became less dependent on average, but every label shared its call with more labels.

C. Recovered Sensitivity: Ignoring Calls Can Understate Uncertainty

To determine whether the observed covariance made 95% confidence intervals miss their target more often than expected, we examined the coverage results in Figure 4, with exact binomial intervals. For Granite/IMDb, the item-level method covered the target in only 77.8% of replicates, while the observed variance was 2.71 times the value that method assumed. The call-aware t interval instead reached 93.0% coverage. Qwen/Civil Comments showed the same pattern, improving from 88.0% to 96.0%.

Mistral/MultiNLI had milder undercoverage (90.8%) and improved to 95.3% with the call-aware interval. Llama/Civil Comments showed no item-level undercoverage. Whole-call bootstrap coverage was 91.8%, 95.3%, 94.8%, and 94.0% for Granite, Qwen, Mistral, and Llama, respectively. Wilson-score intervals reproduced the item-level result, so the undercoverage did not result from using the Wald formula near a probability boundary. Undercoverage varied across cells, but the risk created by positive dependence cannot be diagnosed from the number of labeled rows alone.

V. Discussion

The unit that generates an annotation can also create statistical dependence. A joint prompt may establish a temporary decision rule, interpretation, or output pattern that affects several labels at once. Exact-context controls show that longer prompts or neighboring documents alone do not explain the observed covariance. We therefore analyze each joint call as a cluster while leaving the causal mechanism unresolved.

The batch-size result separates dependence between pairs from the total information loss within a call. Although dependence remained detectable even in small batches, it decreased across the tested range in three selected cells and increased in one. The design effect still rose because the number of other labels sharing each call grew faster than pairwise dependence declined. Reducing batch size can reduce this penalty, but the remaining labels may still be dependent.

This distinction matters when estimating means or proportions from batched labels. Positive covariance within a call can make formulas that assume independent rows underestimate variance. Researchers should retain original and parent call IDs, attempt and recovery status, model and protocol versions, decoding settings, batch size, and output position. These records permit call-aware analysis and auditing. Call-level t intervals, cluster-robust standard errors, or whole-call bootstrap resampling can account for call clustering [4], [16], [17]. The choice depends on the number and structure of independent calls, and studies with few calls require small-sample methods. Sources, users, conversations, or documents may add other forms of clustering that the analysis must also address. These methods change estimated uncertainty, not label accuracy. Effects on other downstream estimates remain untested.

The study has four limitations. First, it evaluates four open-weight models near the 7–8B scale and four English classification tasks, so proprietary, larger, multilingual, generative, and subgroup-specific behavior may differ. Second, the batch-size experiment uses four different cells and sizes 2–32, so it cannot

separate model-family effects from task effects or establish a universal scaling law. Third, the matched control identifies the combined effect of joint annotation but cannot determine whether multi-target instructions, shared-response decoding, or another component produces the dependence. Fourth, the repeated-sampling study required post-execution recovery of 1,095 calls with a modified output format and response-length limit. The unrepaired subsets gave similar item-level coverage: 78.9% for Granite/IMDb (105/133 replicates) and 88.4% for Qwen/Civil Comments, compared with recovered values of 77.8% and 88.0%. These unrepaired subsets were too small for the prespecified decision rule, so the coverage result remains secondary sensitivity evidence.

VI. Conclusion

Across four model families and four tasks at batch size 16, the primary grid showed that errors occurred together more often when labels shared a call. Although dependence between pairs decreased on average in the descriptive four-cell batch-size analysis, the total variance penalty increased. In the recovered sensitivity, confidence intervals that treated 512 labels as independent missed their target too often in three cells, whereas intervals based on the 32 call means came much closer to their intended 95% coverage. Studies using batched labels should record the call and protocol that produced each label, report dependence separately from accuracy, and account for every source of grouping in the analysis.

Acknowledgment: AI Disclosure

OpenAI Codex with GPT-5.6 Sol and GPT-6 Astra assisted with manuscript editing and the development and debugging of analysis code. The author formulated the research question, designed and conducted the study, made all methodological and interpretive decisions, verified the AI-assisted edits and code, and takes full responsibility for the paper and its conclusions.

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